EE4-66
EE9-SO28

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2016

MSc and EEE PART IV: MEng and ACGI

TOPICS IN LARGE DIMENSIONAL DATA PROCESSING

Tuesday, 10 May 10:00 am

Time allowed: 3:00 hours

Corrected copy

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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EE4-66 Topics in Large Dimensional Data Processing

Instructions for Candidates

Answer all four questions. Each question carries 20 marks.

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1. (Convex Optimisation)

(a) Consider a constrained optimization problem of the form

$$\begin{aligned} &\min_{\boldsymbol{x}} f(\boldsymbol{x}) \\ &\text{subject to } h_i(\boldsymbol{x}) \leq 0,\ i = 1, \cdots, m, \\ &\ell_j(\boldsymbol{x}) = 0,\ j = 1, \ldots, r. \end{aligned} \tag{1}$$

State the conditions under which the above optimization problem is convex. [3]

(b) Write the Lagrange dual function corresponding to the optimization problem (1). [2]

(c) Give the KKT conditions for the optimal $\boldsymbol{x}$ that solve (1). [2]

(d) Let $\boldsymbol{Q} \succeq 0$. Use the KKT conditions to solve

$$\min_{\boldsymbol{x}} \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{c}^T\boldsymbol{x} \text{ subject to } \boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}.$$

[3]

(e) The epigraph of a function f is defined as

$$\text{epi}f = \{(\boldsymbol{x}, t) \mid \boldsymbol{x} \in \text{domain}(f),\ f(\boldsymbol{x}) \leq t\}.$$

Sketch the epigraph of $f = x^2$. [2]

(f) Prove that the epigraph of a convex function $f(\boldsymbol{x})$ is convex. [3]

(g) Let $f_1, f_2 : \mathbb{R}^n \to \mathbb{R}$ be convex functions. Define their pointwise maximum f by $f(\boldsymbol{x}) = \max\{f_1(\boldsymbol{x}), f_2(\boldsymbol{x})\}$. Show that f is also convex. [3]

(h) Let $f_1 = x^2$ and $f_2 = (x-1)^2$. Sketch the epigraph of $f(x) = \max\{f_1(x), f_2(x)\}$. [2]

2. (Matrix Property) Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ contain unit ℓ_2-norm columns and $m < n$. Let $\boldsymbol{x}_0 \in \mathbb{R}^n$ be S-sparse. Denote its support by $\mathcal{T}_0$. Let

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{e}. \tag{2}$$

In this problem, we assume that $\|\boldsymbol{e}\|_2 \leq \epsilon$.

(a) What is the definition of spark $(\boldsymbol{A})$? [2]

i Suppose that $\boldsymbol{A}$ is generated from the Gaussian random matrix ensemble. What is the most likely value of spark $(\boldsymbol{A})$. (No proof is required.) [1]

(b) What is the definition of mutual coherence μ of a matrix $\boldsymbol{A}$? [2]

(c) What is the definition that $\boldsymbol{A}$ satisfies the Restricted Isometry Property (RIP) for $S \leq m$ and $\delta \in (0, 1)$? What is the definition of Restricted Isometry Constant (RIC) δ_S? [4]

(d) Suppose that the matrix $\boldsymbol{A}$ satisfies RIP with RIC δ_S.

i Let $\mathcal{T} \subset [n]$ and $|\mathcal{T}| = S$. Let $\lambda(\boldsymbol{A}_\mathcal{T})$ denote a singular value of the matrix $\boldsymbol{A}_\mathcal{T}$. Find the range of $\lambda(\boldsymbol{A}_\mathcal{T})$. [2]

ii Suppose that the support $\mathcal{T}_0$ of $\boldsymbol{x}_0$ is known a priori. Give the formula to compute the estimate $\hat{\boldsymbol{x}}$ of $\boldsymbol{x}_0$ from $\boldsymbol{y}$ to minimize the ℓ_2-error $\|\boldsymbol{x}_0 - \hat{\boldsymbol{x}}\|_2$. [2]

iii Use δ_S to construct lower and upper bounds on the ℓ_2-error $\|\boldsymbol{x}_0 - \hat{\boldsymbol{x}}\|_2$. [4]

(e) Suppose that the matrix $\boldsymbol{A}$ satisfies RIP with RIC δ_{2S}. Let $\mathcal{I}, \mathcal{J} \subset [n]$ be such that $\mathcal{I} \bigcap \mathcal{J} = \phi$ and $|\mathcal{I} \bigcup \mathcal{J}| \leq 2S$. For any $\boldsymbol{a} \in \mathbb{R}^{\mathcal{I}}$ and $\boldsymbol{b} \in \mathbb{R}^{|\mathcal{J}|}$ such that $\|\boldsymbol{a}\|_2 = \|\boldsymbol{b}\|_2 = 1$, prove that $|\langle \boldsymbol{A}_\mathcal{I}\boldsymbol{a}, \boldsymbol{A}_\mathcal{J}\boldsymbol{b} \rangle| \leq \delta_{2S}$. (Hint: consider the vectors $[\boldsymbol{A}_\mathcal{I}\boldsymbol{A}_\mathcal{J}] \begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \end{bmatrix}$ and $[\boldsymbol{A}_\mathcal{I}\boldsymbol{A}_\mathcal{J}] \begin{bmatrix} \boldsymbol{a} \\ -\boldsymbol{b} \end{bmatrix}$.) [3]

3. (AMP)

Consider the soft thresholding function

$$\hat{x} = \eta(y;\theta) := \operatorname{sign}(y) \cdot \max(0, |y| - \theta) = \begin{cases} y - \theta & \text{if } y > \theta, \\ 0 & \text{if } -\theta \leq y \leq \theta. \\ y + \theta & \text{if } y < -\theta. \end{cases}$$

(a) Consider the system $Y = X + W$, where

$$p_X(x) = \frac{\epsilon}{2}\delta_{-\infty} + (1-\epsilon)\,\delta_0 + \frac{\epsilon}{2}\delta_{+\infty} \text{ and } W \sim \mathcal{N}(0, \sigma^2).$$

Let $\theta = \alpha\sigma^2$ and define

$$\operatorname{mse}(\sigma^2; \alpha) = \mathrm{E}_{X,W}\left[(\eta(Y;\theta) - X)^2\right].$$

i Prove that $\operatorname{mse}(\sigma^2;\alpha) = \operatorname{mse}(1;\alpha)\,\sigma^2$. [3]

ii The quantity of $\operatorname{mse}(1;\alpha)$ can be computed as follows. Let $\phi(\cdot)$ and $\Phi(\cdot)$ are the pdf and cdf of $W \sim \mathcal{N}(0,1)$.

A. Compute $\mathrm{E}_W\left[(\eta(Y;\alpha) - X)^2 \,|x = \infty\right]$. [3]

B. Compute $\mathrm{E}_W\left[(\eta(Y;\alpha) - X)^2 \,||w| \leq \alpha, x = 0\right]$. [2]

C. Compute $\mathrm{E}_W[\alpha | w > \alpha]$. [2]

D. Let $c_1 := \mathrm{E}_W\left[(\eta(Y;\alpha) - X)^2 \,|x \neq 0\right]$, $c_2 := \mathrm{E}_W\left[(\eta(Y;\alpha) - X)^2 \,|x = 0\right]$. Compute $\operatorname{mse}(1,\alpha)$. [2]

(b) Let $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{w}$, where $\boldsymbol{w} \sim \mathcal{N}(0, \sigma^2\boldsymbol{I})$. Let $\boldsymbol{x}^t$ be an estimate of $\boldsymbol{x}_0$. Let

$$\boldsymbol{x}^{t+1} = \eta\left(\boldsymbol{x}^t + \boldsymbol{A}^T\left(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^t\right); \theta\right).$$

Define $\tau_t := \frac{1}{N}\left\|\boldsymbol{x}^t - \boldsymbol{x}_0\right\|_2^2$ and $\tau_{t+1} := \frac{1}{N}\left\|\boldsymbol{x}^{t+1} - \boldsymbol{x}_0\right\|_2^2$.

In the following, assume that $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ is a Gaussian random matrix of which the entries $A_{i,j} \sim \mathcal{N}\left(0, \frac{1}{m}\right)$ are i.i.d. Let $\boldsymbol{a}_i$ be the i-th column of $\boldsymbol{A}$.

i Show that $\boldsymbol{x}^t + \boldsymbol{A}^T\left(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^t\right) = \boldsymbol{x}_0 + \left(\boldsymbol{A}^T\boldsymbol{A} - \boldsymbol{I}\right)\left(\boldsymbol{x}_0 - \boldsymbol{x}^t\right) + \boldsymbol{A}^T\boldsymbol{w}$. [2]

ii Given $\boldsymbol{w}$, each entry of $\boldsymbol{A}^T\boldsymbol{w}$ is Gaussian with zero mean. Show that its variance is $\frac{1}{m}\|\boldsymbol{w}\|_2^2$. [2]

Note that $\frac{1}{m}\|\boldsymbol{w}\|_2^2 \to \sigma^2$ as $m \to \infty$. It holds that each entry of $\boldsymbol{A}^T\boldsymbol{w}$ is $\mathcal{N}(0, \sigma^2)$ approximately.

iii Assume that the entries in each row of $\boldsymbol{B} := \boldsymbol{A}^T\boldsymbol{A} - \boldsymbol{I}$ are approximately i.i.d. with distribution $\mathcal{N}\left(0, \frac{1}{m}\right)$. It can be shown that each entry of $\left(\boldsymbol{A}^T\boldsymbol{A} - \boldsymbol{I}\right)\left(\boldsymbol{x}_0 - \boldsymbol{x}^t\right)$ is $\mathcal{N}(0, c)$. Find the value of c in terms of τ_t. [2]

iv Let $m/n = \delta$. Find the state evolution relationship between τ_t and τ_{t+1}. [2]

4. (Greedy Algorithms) Consider the linear system

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{e}, \tag{3}$$

where $\boldsymbol{x}_0$ is S-sparse. Define the hard thresholding function $H_s(\boldsymbol{a})$ by setting all but the largest (in magnitude) s elements of $\boldsymbol{a}$ to zero. Let $\operatorname{supp}(\boldsymbol{a}) := \{i : a_i \neq 0\}$.

(a) Write down the Orthogonal Matching Pursuit (OMP) algorithm to solve the sparse linear inverse problem (3). [3]

Let μ be the mutual coherence of $\boldsymbol{A}$. The Cauchy-Schwartz inequality, i.e., $\|\boldsymbol{z}\|_1 \leq \sqrt{\ell}\,\|\boldsymbol{z}\|_2$ for all $\boldsymbol{z} \in \mathbb{R}^\ell$, will be useful.

i Find an upper bound on $\max_{i \notin \mathcal{T}_0} |\langle \boldsymbol{a}_i, \boldsymbol{y} \rangle|$ in terms of μ and $\|\boldsymbol{x}_0\|_2$. [3]

ii Find a lower bound on $\max_{i \in \mathcal{T}_0} |\langle \boldsymbol{a}_i, \boldsymbol{y} \rangle|$ in terms of μ and $\|\boldsymbol{x}_0\|_2$. [3]

iii Find the condition of μ such that in the first iteration of OMP the chosen index $i_1 \in \mathcal{T}$. Prove your claim. [3]

(b) Write down the Iterative Hard Thresholding (IHT) algorithm to solve the sparse linear inverse problem (2). [3]

i Let $\boldsymbol{r}^k := \boldsymbol{x}_0 - \boldsymbol{x}^k$ ($\boldsymbol{r}^0 = \boldsymbol{x}_0$). It has been shown that if $\delta_{3S} < 1/\sqrt{32}$, the estimation error of IHT satisfies

$$\left\|\boldsymbol{r}^{k+1}\right\|_2 \leq 0.5 \left\|\boldsymbol{r}^k\right\|_2 + 2.5 \|\boldsymbol{w}\|_2 .$$

Use this fact to derive proper constants c_k and c' such that $\left\|\boldsymbol{r}^k\right\|_2 \leq c_k \|\boldsymbol{x}_0\|_2 + c' \|\boldsymbol{w}\|_2$. [3]

ii If $\boldsymbol{w} = \boldsymbol{0}$, find the number of iterations for IHT to guarantee $\left\|\boldsymbol{r}^k\right\|_2 \leq 10^{-6}$. [2]